



Vision-Based Object Detection and Proportional Navigation for UAS Collision Avoidance

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This paper focuses on the development of vision-based guidance algorithms to enable an unmanned aerial system (UAS) to sense and avoid other vehicles or stationary obstacles within its operational environment. Image processing algorithms are developed and implemented with an emphasis on the detection and tracking of other unmanned aerial vehicles. Other algorithms considered in this work include edge detection and the Hough transform for power line detection as well as feature point tracking algorithms for the detection of stationary objects. Guidance laws based on proportional navigation are employed to generate collision avoidance maneuvers when the measured line-of-sight rates to another vehicle or object are small, indicating a potential risk of collision. Examples are provided in which template matching and centroiding algorithms are employed to detect, track, and measure line-of-sight rates of a UAV from onboard imagery obtained from another UAV. Representative results are also presented in which feature point tracking algorithms are integrated with the proportional navigation guidance algorithms to enable a simulated UAV to sense and avoid buildings within a virtual urban environment.

I. Introduction

Autonomous navigation of unmanned aerial systems (UAS) through complex urban environments requires the ability to sense and avoid a diverse range of stationary and dynamic obstacles. This paper extends previous work [1 – 3], in which guidance strategies for collision avoidance were developed based on the principle of proportional navigation, to consider image processing algorithms that can be employed to detect and track various collision threats within complex environments.

The image processing techniques investigated in this paper include methods for detecting another UAV in flight such as template matching and feature detection in combination with clustering. In addition, methods for detecting linear features, such as power lines, using techniques such as the application of a Canny edge detector in conjunction with the Hough transform. These cases are representative of some of the challenging objects that a UAV might need to sense and avoid to successfully navigate through an urban environment.

This paper develops image processing algorithms applied to the detection of objects such as a UAV in flight and power lines. These algorithms can be used to detect and track various collision threats, providing information to proportional navigation based guidance laws. These guidance laws provide a UAS with commands to execute a collision avoidance maneuver when the measured line-of-sight rates to a particular object are small, a condition that would indicate a potential collision. The paper provides examples in which onboard UAV imagery is post processed to identify and track the centroid of another UAV, information that can be used to compute line-of-sight rates and guidance commands if there is a potential collision. Simulation examples are also presented in which feature point detection and tracking algorithms are used to generate guidance commands to enable a UAV to avoid obstacles within a virtual urban environment.

II. Image Processing Algorithms for Object Detection

Two methods of identifying another UAV in flight have been investigated and implemented. First, a template matching method was investigated. A side view of a DJI *Inspire* was downloaded from DJI's promotional material and used as the template. The drone was against a plain white background. Three test images were used in which it was attempted to identify another drone in the image that matches the template. The method used was to compute the cross-correlation matrix between the template and the image. The normalized cross-correlation of two matrices provides values between $(-1.0, 1.0)$ and is related to the covariance matrix where the output has rows of observations and columns of variables. The regions of maximum correlation (above a given threshold) were identified and used to determine the area of the bounding box for the image. This method was first discussed by J.P. Lewis in "Fast Template Matching", *Vision Interface*, p120-123, 1995 [4]. This method is additionally able to be implemented on a GPU for fast, real-time processing. Figure 1 presents results demonstrating that the object was found in images with varied viewing angles and sizes.



Figure 1: Successful identification of DJI *Inspire* from in-flight images.

The second method of identifying drones in flight was a clustering of features. The Features from Accelerated Segment Test (FAST) feature detection algorithm [5] was used because of its ease of use and its existing implementation in MATLAB. This well-known algorithm identifies the location of features from an optical image. Additionally, the FAST algorithm has been optimized for use on a GPU processor, and therefore, it is suitable for an onboard flight implementation.

Once FAST features were found, clusters were identified as features that were within a threshold number of pixels from other features, with the idea that an object of interest would have numerous features. Once the clusters of features were identified, a window was created around the object. From the windowed image, a centroid was produced based on the intensity. Figure 2 shows the identified features on a DJI *Inspire* and Figure 3 shows identified features and centroids of two different drones (DJI *Inspire* and *Phantom*) in flight.

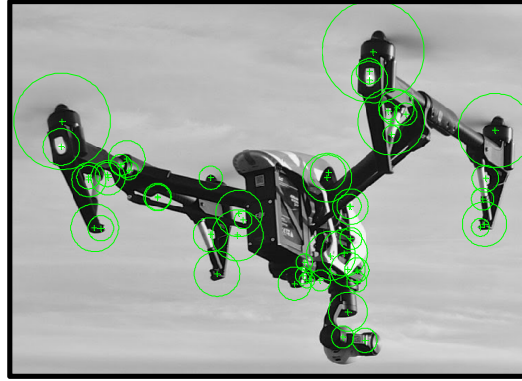


Figure 2: Identified FAST features.

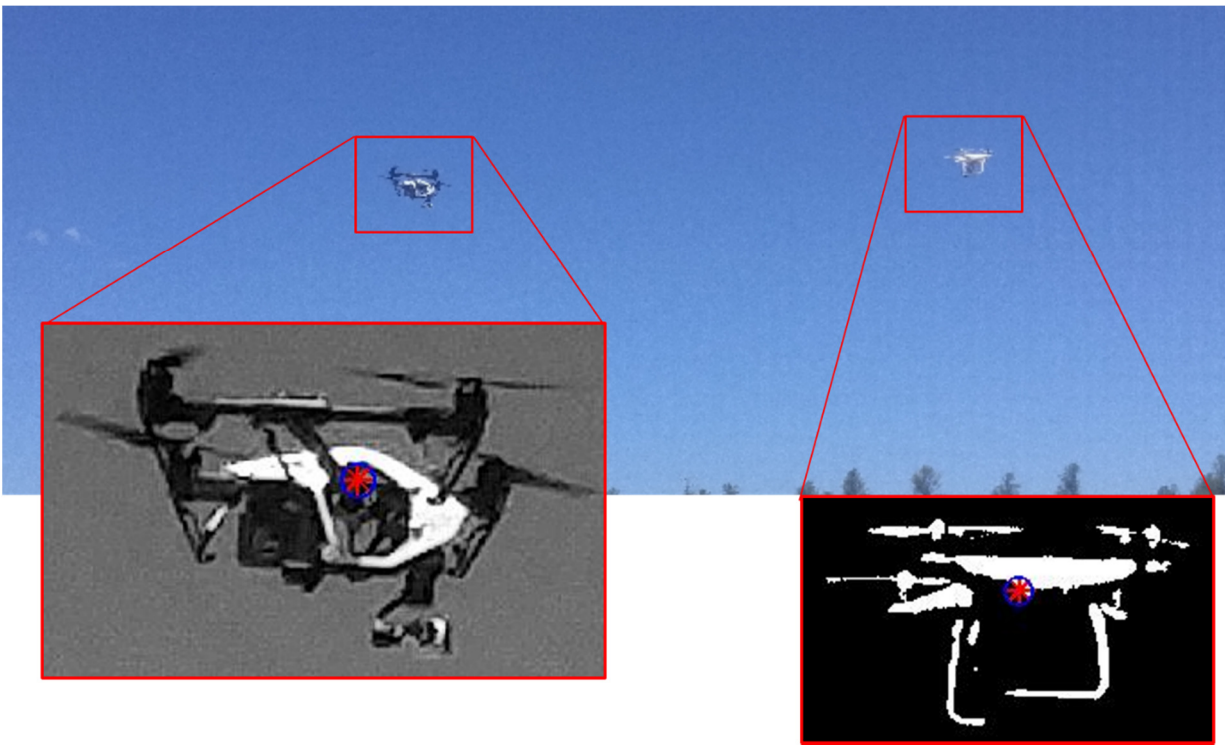


Figure 3: Centroids produced after windowed via clusters of features method.

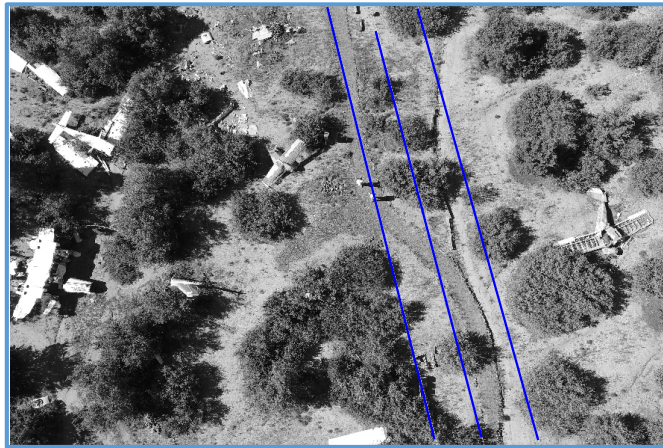
Metrics on detection range and object size can be defined as a function of geometry and the sensor. Minimum bounds can also be identified for template matching ability and over multiple templates. While metrics on moving targets can be defined, but only individual images have been considered in this work.

Image processing methods were also developed and implemented to identify power lines in optical images from a UAV.

The images were analyzed in MATLAB using both algorithms in the Image Processing Toolbox and custom image processing tools developed by the authors. First, the image was converted from color to grayscale by eliminating the image hue and saturation information while retaining the luminance information. The weighted sum of the red, green, and blue channels was standard for the Bayer filter to intensity conversion.



Next, the edges were located in the image using the Canny method [6], which looks for local maxima of the gradient of the image (through the derivative of a Gaussian filter). The Canny method uses two different thresholds to detect strong and weak edges and includes the weak edges as output only when they are connected with strong edges. This method is therefore more robust to noise and more likely to detect true weak edges.



Finally, the line segments are identified through the Hough transform [7], which uses a parametric representation of a line.

Figure 4: Captured image (top), and identified power lines (bottom).

Only lines that were longer than the smallest dimension of the image were extracted for the final estimate, as the Hough transform identified 75 total lines in the image provided in Figure 4. This restriction means that the power lines to be identified must traverse the entire image, and not only one corner, for example. Identifying power lines in a fraction of the image (e.g., through a corner) is an ongoing research topic.

III. Proportional Navigation Based Collision Avoidance

Reactive obstacle avoidance algorithms have been developed by the authors [1 – 3] to address obstacles that are not properly accounted for in a terrain map. These would include dynamic obstacles, such as other vehicles, as well as stationary obstacles that suddenly appear in front of the UAV, such as the case where the UAV turns a corner and suddenly encounters an obstacle. The avoidance guidance laws are based on the principle of proportional navigation, which states that an object is on a collision course if the line-of-sight (LOS) angular rates to that object remain constant. Equivalently, if an obstacle appears constant in the field of view of a UAV camera, the UAV is on a collision course with that object. The pronav-based collision avoidance laws seek to ensure that the LOS rates to detected objects in the scene do not approach zero, which would be indicative of a potential collision.

The pronav-based guidance law uses the rates of change of the LOS angles, $\dot{\chi}_{LOS}$ and $\dot{\gamma}_{LOS}$, which can be derived from the change in position of the target object projected on the image plane. These horizontal and vertical LOS angles, which are illustrated in Figure 5, can be derived as

$$\chi_{LOS} = \tan^{-1} \left(\frac{\hat{y}_T^E}{\hat{x}_T^E} \right)$$

$$\gamma_{LOS} = \sin^{-1} \left(\frac{\hat{z}_T^E}{\sqrt{(\hat{x}_T^E)^2 + (\hat{y}_T^E)^2}} \right)$$

where the target location relative to the camera must first be converted from camera to inertial coordinates. The lateral and vertical line-of-sight rates can then be estimated using finite differences as:

$$\dot{\chi}_{LOS}(t_k) = \frac{\chi_{LOS}(t_k) - \chi_{LOS}(t_{k-1})}{\Delta t}$$

$$\dot{\gamma}_{LOS}(t_k) = \frac{\gamma_{LOS}(t_k) - \gamma_{LOS}(t_{k-1})}{\Delta t}$$

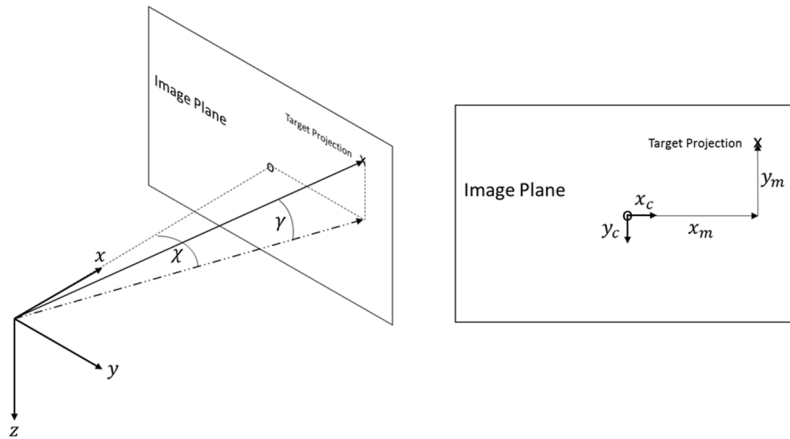


Figure 5: Line-of-Sight (LOS) Angles Derived from Camera Measurements.

A pronav-based avoidance law was implemented based on the measured LOS rates to multiple moving targets in the scene. For a single target case, the guidance law commands a change in heading if the LOS rates to the object are close to zero; otherwise, no action is taken and the UAV continues with its mission. For multiple objects, the LOS rates are measured for each object. If more than one object has LOS rates that are close to zero, the UAV changes heading based on the object with the smallest LOS rates. This most imminent threat approach is demonstrated in Figure 6, which depicts a scenario in which a UAV is flying due north towards a desired waypoint but encounters two other UAVs moving in across its path. In this scenario, the UAV would collide with one or both dynamic obstacles if no action was taken. With the active pronav avoidance law, the UAV deviates to move slightly behind the flight path of both dynamic obstacles and then proceeds to the specified waypoint.

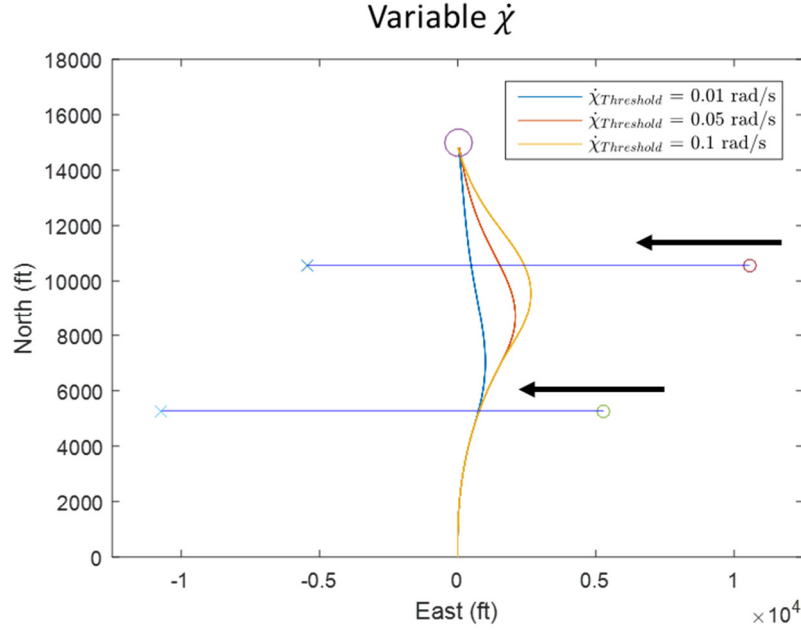


Figure 6: UAV ProNav-Based Avoidance of Two Dynamic Obstacles using the Most Imminent Threat Approach.

An alternative, cost-function based approach was developed to account for all obstacles rather than only the one that poses the most imminent threat. The proposed guidance law is represented by the cost function below, which incorporates terms to attract the UAV to the waypoint as well as terms to drive the UAV away from headings that could lead to a collision, to solve for a resultant commanded heading, ψ_{com} .

$$J = W_{WP}(\psi_{com} - \psi_{WP})^2 + \sum_{i=1}^N \frac{W_O}{(\psi_{com} - \psi_{O_i})^2}$$

The first part of the equation, $W_{WP}(\psi_{obj} - \psi_{WP})^2$, represents the objective to intercept the waypoint (recalling the original pronav intercept implementation), while the second part incorporates the need for an avoidance maneuver, taking into account all obstacles in view with the potential to be a threat. The weighted values, W_{WP} and W_O , allow for tuning of the cost function to appropriately gauge its effectiveness in a multi-engagement scenario. ψ_{WP} is the commanded heading required to ensure an intercept with the defined waypoint and ψ_{O_i} is the commanded heading for an intercept with the i_{th} obstacle, such that:

$$\psi_{WP} = \psi_{A/C} + N_H \dot{\chi}_{WP} \Delta t_c$$

and

$$\psi_{O_i} = \psi_{A/C} + N_H \dot{\chi}_{O_i} \Delta t_c$$

Therefore, minimizing the cost function yields a commanded heading that balance the objective of intercepting the desired waypoint while avoiding headings that will intercept the obstacles.

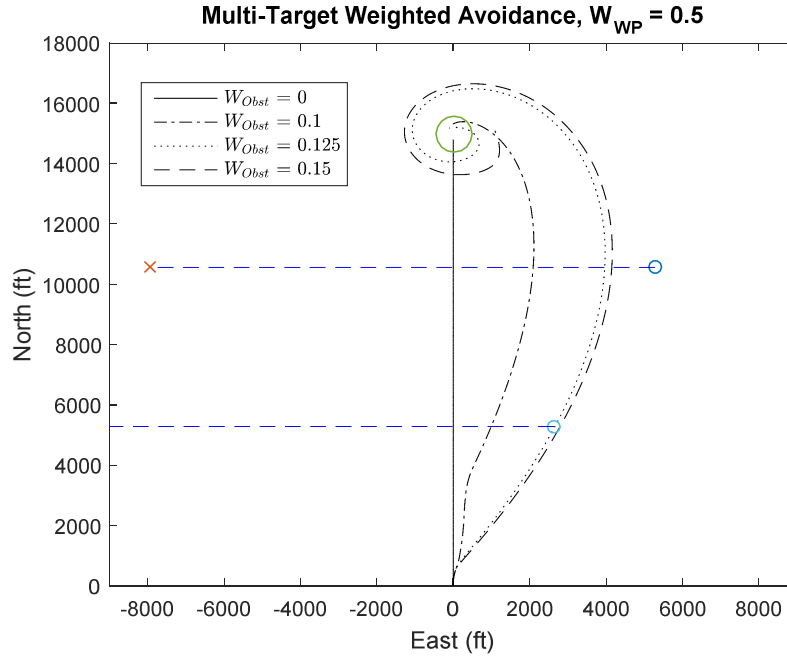


Figure 7: UAV ProNav-Based Avoidance of Two Dynamic Obstacles using a Cost Function Approach.

An example of the cost function based approach is provided in Figure 7 for the same scenario shown in Figure 6. The different avoidance paths represent varying weightings in the cost function. This cost function based approach will be investigated further. Simulations will be performed in the virtual environment to evaluate the performance of the algorithm for the case in which there are numerous obstacles in the scene.

IV. Dynamic Obstacle Identification from UAV Imagery

A template matching based algorithm for identifying known types of dynamic obstacles in flight has been implemented to detect a quadcopter UAV crossing the flight path of the primary vehicle using both a visible light and infrared camera. The goal of the algorithm is to generate line-of-sight (LOS) unit vectors from the primary UAS' camera frame to the detected object, which can then be used to find the lateral and vertical line-of-sight rates for use in the control navigation architecture. For the purpose of testing the pronav performance through a series of images, the type dynamic obstacle is assumed to be known (a DJI quadcopter in flight). Figure 8 shows an example of the path followed by an identified UAS (DJI Inspire) through a series of infrared images.

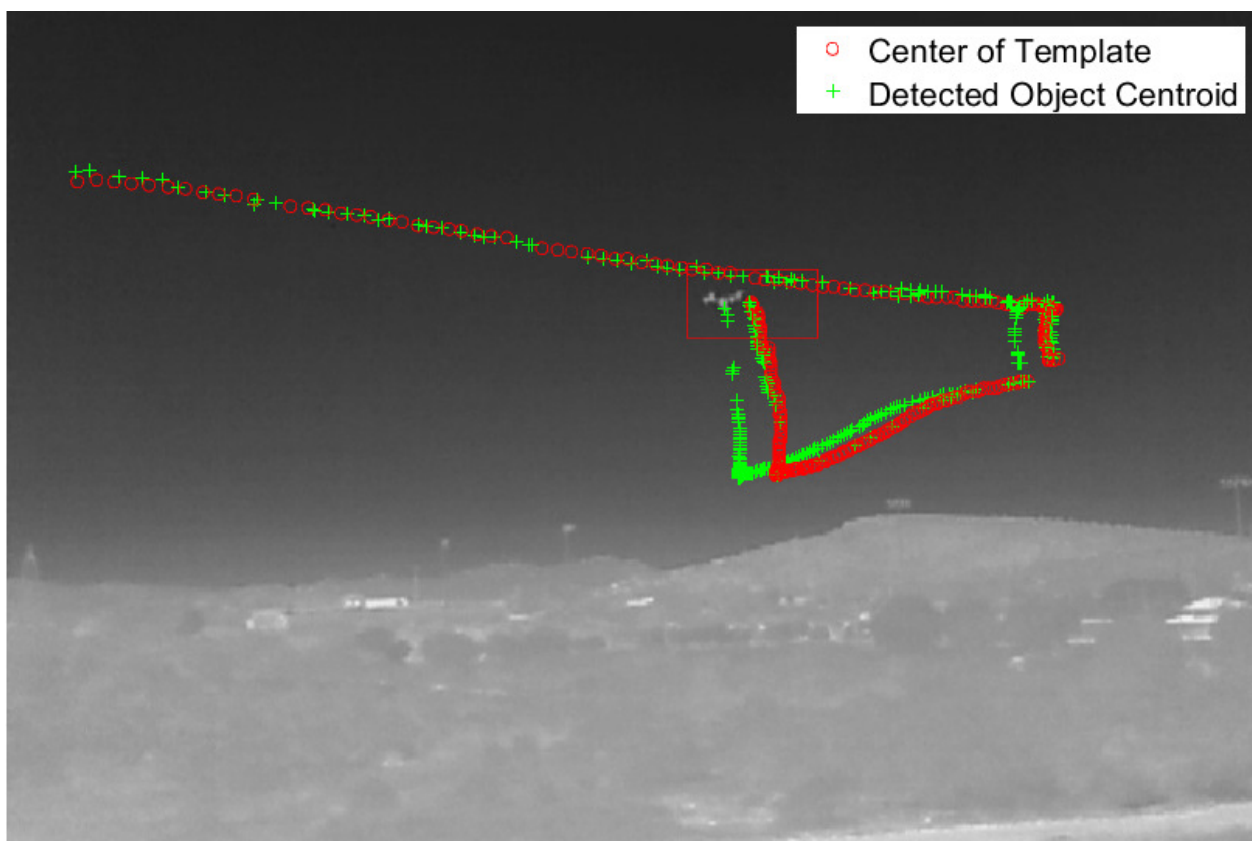


Figure 8: Detected object centroids time sequence infrared example

A template image of a known object (quadcopter UAS in this example) of interest initializes the algorithm. Each image is then processed, by converting it to grayscale and/or binary, to maximize the likelihood of object detection. MATLAB's built-in template matching tool attempts to find a match for the object of interest within each data image using a normalized cross-correlation algorithm, which is invariant of scale. Once the maximum correlation is identified, a sub-window identifying the object is defined in the original image. The sub-window containing the object is partitioned from the full image for further processing.

An intensity-based center-of-mass centroiding algorithm is then applied to the sub-window to locate the true centroid of the object of interest. As seen in Figure 9, the sub-window often contains multiple centroids for non-contiguous regions of a detected object. The challenge then is to properly find the actual centroid of the object from this collection of points. Once a total centroid is found, the algorithm continues to the next data image. If criteria for updating the template image are met, the true centroid is utilized to update the template image by choosing a new sub-image centered around the true centroid. Real-time updates of the template allow frame-to-frame detection of objects changing due to movement, orientation, or illumination differences (within frame-rate limitations). An overview of the algorithm is shown in Figure 10.

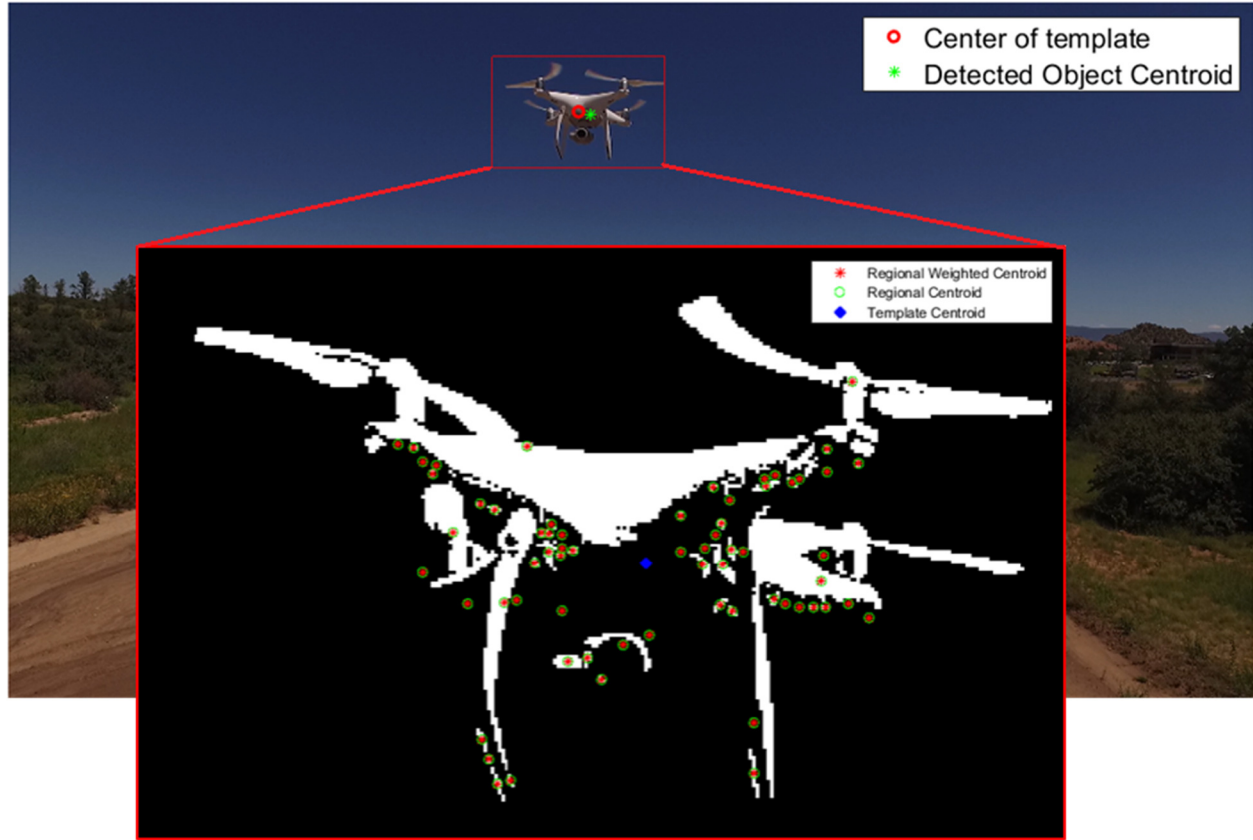


Figure 9: Sub-window with centroid points plotted, overall centroid of the sub-window determined by finding weighted center of mass (intensity based)

Once centroids are obtained the next step is to transform this data into usable commands for the proportional navigation collision avoidance control laws. Camera parameters are assumed to be known and a pinhole camera model is used to transform the 2D centroid points on the image plane into LOS unit vectors in the camera's coordinate frame. Image distortion corrections will not be considered in this paper. We will assume that our UAS camera system performs image distortion correction before images are sent to the algorithm.

Pixel coordinates are related to camera frame coordinates through the camera calibration matrix K . A rotation matrix R is also employed to move to a standard camera frame orientation. A rotation matrix R_{cam} is required since the pixel axes are (x – right, y – down), and image axes are (x – up, y – right). Given a camera systems specifications, (i.e. focal length, image dimensions, sensor size and FOV), we can form the camera calibration matrix which moves the origin of the pixel axes from the upper left corner to the center, and accounts for the actual size of the sensor dimensions:

$$K = \begin{bmatrix} ratio_H & 0 & Offset_x \\ 0 & ratio_V & Offset_y \\ 0 & 0 & f \end{bmatrix}$$

where the terms $ratio_H$ and $ratio_V$, transform the pixel coordinate sizes in the horizontal and vertical directions into camera frame sizes (mm). We also wish to define a rotation matrix R_{cam} to change the orientation of our coordinate frame axes to our camera frame.

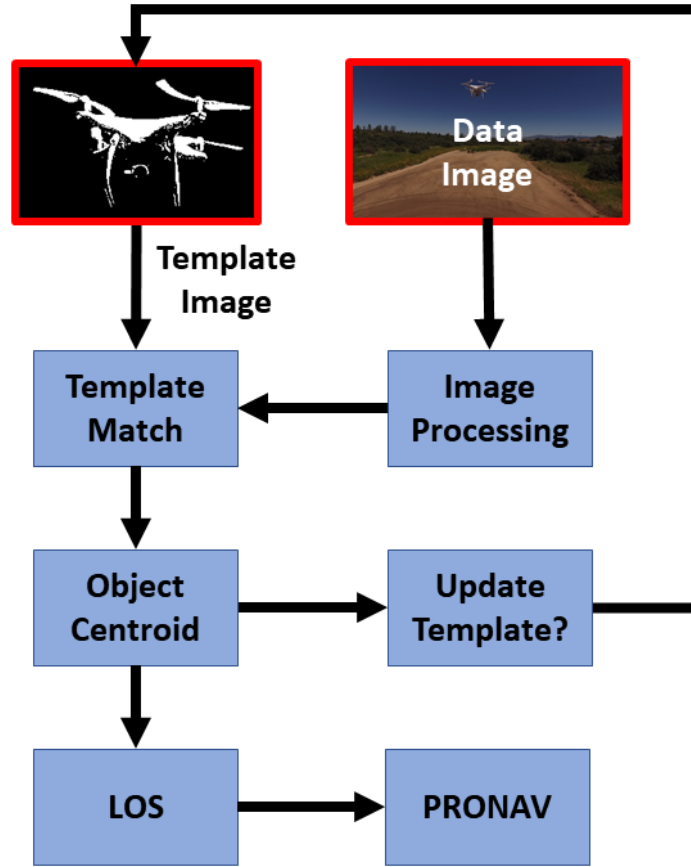


Figure 10: Dynamic Object Identification Algorithm Overview

$$R_{cam} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Once we have the camera calibration matrix and the rotation, we can transform the centroid location from the pixel frame to the camera frame through the following:

$$\begin{bmatrix} x_p \\ y_p \\ f \end{bmatrix} = KR_{cam} \begin{bmatrix} x_c \\ y_c \\ f \end{bmatrix}$$

$$\begin{bmatrix} x_c \\ y_c \\ f \end{bmatrix} = (KR)^{-1} \begin{bmatrix} x_p \\ y_p \\ f \end{bmatrix}.$$

We then find line of sight (LOS) unit vectors in the camera frame simply by dividing the set of camera frame coordinates by its' respective norm. That is by taking the camera frame vector, $\vec{x}_c = \begin{bmatrix} x_c \\ y_c \\ f \end{bmatrix}$, and forming the LOS as:

$$\hat{\mathbf{L}} = \frac{\vec{x}_c}{\|\vec{x}_c\|}.$$

The LOS rates can then be estimated via finite differences.

V. Simulation Results from a Virtual Urban Environment

Vision-based proportional navigation simulations were performed using a 6-DOF dynamic model of a UAV operating in a virtual urban environment. MetaVR Virtual Reality Scene Generator (VRSG) software was used to define the virtual urban environment and generate synthetic images representing camera images collected from the simulated UAV. In these simulation studies, the bearings to static obstacles are measured using a Kanade-Lucas-Tomasi (KLT) feature point tracking algorithm [9, 10]. Corners, or intersections of high gradient profiles in the image provide feature points that have the potential and quality to be found and tracked throughout a dynamic scene. The Harris corner detection algorithm identifies a corner, defined when a small area, or window, of pixels in the image exhibits a large intensity change in all directions when the window is shifted from its original location. The KLT tracking algorithm attempts to find these corners in the next image generated and determines the direction and magnitude of the change at subpixel resolution.

Single Obstacle Avoidance

The first scenario places a single building between the UAV and a goal waypoint within the virtual environment, 1,500 *ft* from the UAV. It is important to note that the waypoint is still represented by a point mass, whose LOS angle and rate measurements are derived using the same method of the previous point mass simulations. Figure 11 shows the initial image collected from a simulated camera with total vertical and horizontal view angles of 36° and 60°, respectively.



Figure 11: UAV Camera view of a single building within the virtual environment

This simulation, and the ones to follow, utilize the cost-function-based avoidance law, which is necessary to generate collision avoidance commands based on numerous tracked feature points. Due to the nature of a virtual environment, new criteria implemented for handling tracked feature points, such that both their horizontal and vertical LOS rates must be below a threshold to be considered for cost function minimization, rather than just the horizontal LOS rate being used previously. Figure 12 shows an

instantaneous capture of the camera view point with overlaid feature points selected for the avoidance law to use in determining the UAV maneuver.



Figure 12: Tracked feature points used for single building avoidance

Figure 13 shows the result of the cost function based avoidance implemented within the virtual environment. The left image in this figure is representative of the physical travel and avoidance distances while the image on the right shows a top down view of the path in relation to the virtual environment. The paths shown are considered successful avoidances and resembles those seen in the previous point-mass simulations.

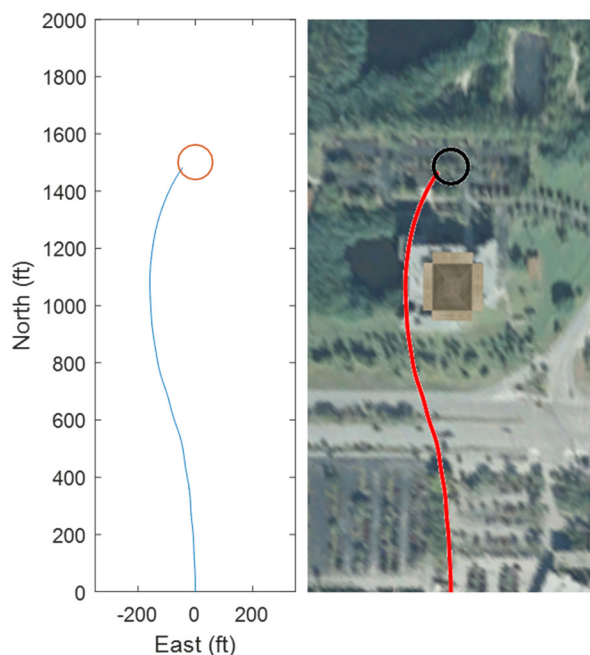


Figure 13: UAV avoidance path for a single building

Figure 14 shows the resulting commanded and actual heading history for this example. There is a slight oscillation in the heading but an overall increase related to the avoidance maneuver around the building,

followed by a decrease in heading while the UAV returns to a waypoint intercept trajectory. The slight oscillation in the later portion of the history reflects on the temporary appearance of feature points detected along the textured ground and horizon line.

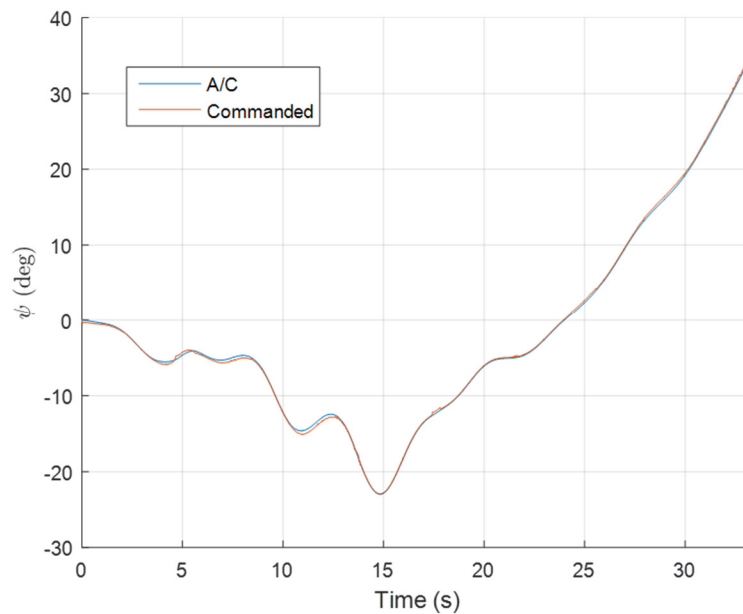


Figure 14: UAV heading history for single building avoidance

Urban Canyon Avoidance

The next scenario considered places the UAV in the center of an urban canyon within the virtual environment. An objective waypoint is placed directly in front of the UAV at 2,000 *ft*, giving the UAV avoidance law the task of navigating the UAV straight through the urban canyon. As mentioned for the scenario of a single building, the waypoint is still represented by a point mass, whose LOS angle and rate measurements are derived using the same method of the previous point mass simulations. Figure 15 shows the initial synthetic camera view captured in the simulation, with total vertical and horizontal view angles of 36° and 60°, respectively. A scene capture with overlaid feature points is shown in Figure 16.

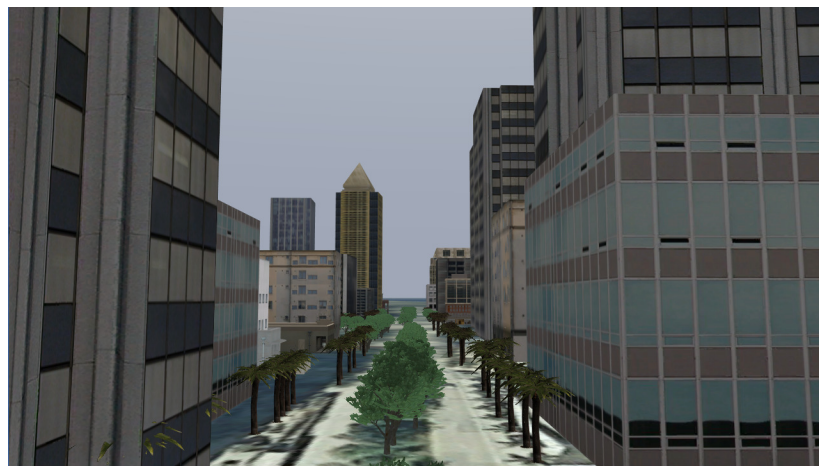


Figure 15: Camera view from UAV within a simulated urban canyon



Figure 16: Feature point overlay of UAV camera view within the urban canyon

The cost function-based avoidance law resulted in the UAV path shown in Figure 17. The path shows some deviations from the center of the canyon based on the location of the feature points detected within the field of view and the number of feature points that have horizontal LOS rates below the threshold for avoidance. The initial deviation to the right indicates that the best course of action for the UAV was to maneuver slightly until more threatening features on the left-hand side of the canyon were no longer a threat, or until the threat balanced out with potential threats on the right-hand side. Later in the path, slight deviations can be seen as the UAV briefly turns towards the left, but returns to a far-right position as before. Finally, the UAV makes a maneuver to intercept the waypoint as the waypoint LOS rate increases, outweighing any potential threats along the sides of the canyon.

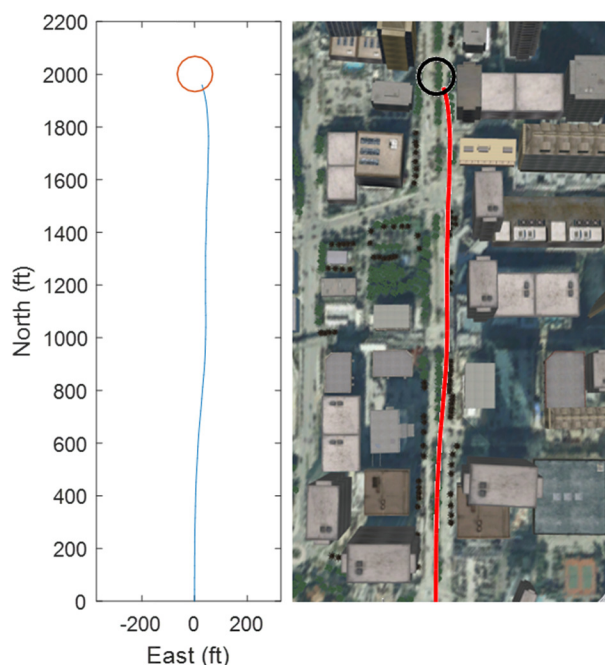


Figure 17: UAV avoidance path for the urban canyon scenario

The UAV heading history for the urban canyon scenario can be seen in Figure 18. Here, the heading trend throughout the entire flight correlates to the flight path shown in Figure 17 and also provides insight

on the not so obvious smaller deviations that the UAV exhibited during the flight. Small oscillations are seen throughout the process of increasing and decreasing the UAV heading, due to the maneuver placing old feature points outside the threatening threshold of LOS rate and placing newer points within that threshold, creating a temporary new heading. This oscillation is similar to that of the single building scenario heading history; however, in this scenario, due to more buildings and potential feature points being present, the effect is more appreciable.

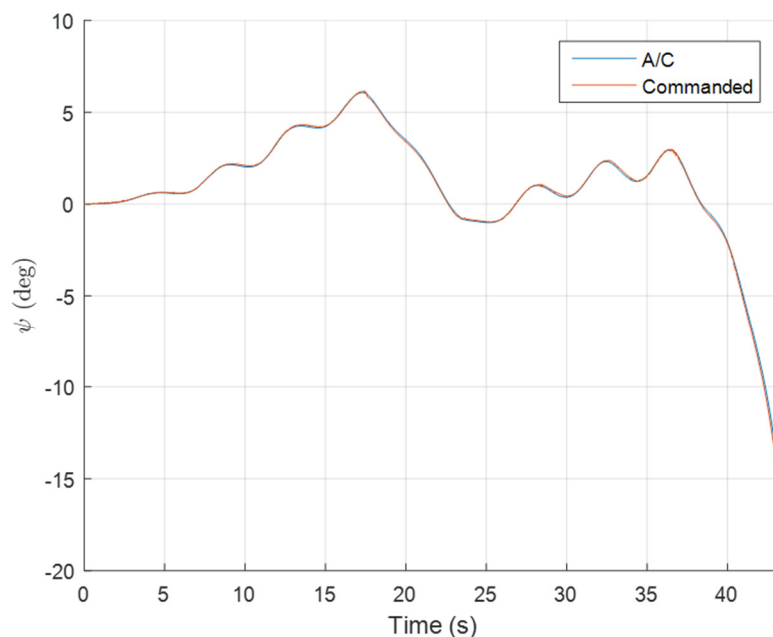


Figure 18: UAV heading history for the urban canyon scenario

VI. Conclusion

This paper has developed and implemented image processing algorithms in order to detect and measure the line-of-sight (LOS) rates to one or more collision threats within the operational environment of a UAS. These image processing algorithms included template matching and centroid algorithms to detect and avoid another UAV in flight as well as edge detection algorithms for power line detection and feature point tracking algorithms for stationary obstacle avoidance. The measured LOS rates are used in proportional navigation based guidance laws to generate avoidance maneuvers when there is a collision threat, as indicated by small LOS rates to an object. Examples were provided using processed UAV imagery of another UAV as well as vision-based UAV simulations within a virtual urban environment.

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